UNIT 1 • LIVING SYSTEMS & HUMAN BIOLOGY

WE ARE HUMANS!

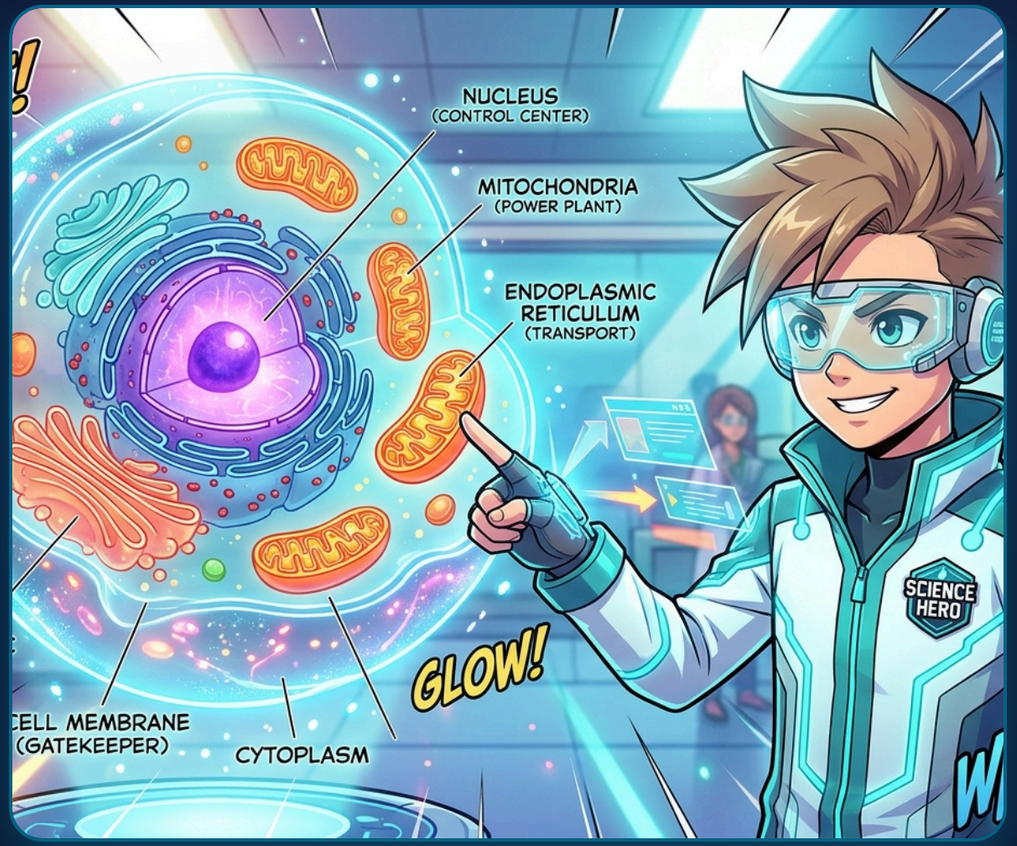
The Secrets of Human Cells, Tissues, Organs & Life Stages

(Los secretos de las células humanas, tejidos, órganos y etapas de la vida)

MISIÓN 1 • BIOLOGY
Cell Architecture

MISIÓN 2 • PHYSIOLOGY
5 Levels of Life

MISIÓN 3 • GROWTH
Human Life Cycle



1. What Makes Us Human?

¿Qué nos diferencia y nos hace únicos como seres humanos?

🚶 Bipedalism

Bipedestación (caminar sobre dos piernas)

Humans stand upright and walk on two feet. This frees our hands to hold tools, craft objects, and explore the world!

👍 Opposable Thumbs

Pulgares oponibles

Our thumbs can touch every other finger on the same hand. This gives humans incredible precision to write, build, and play!

🧠 Complex Brain

Cerebro muy desarrollado

Our highly developed cerebrum allows reasoning, planning, complex emotions, scientific discoveries, and artistic creation.

🗣️ Advanced Language

Lenguaje abstracto y comunicación

We communicate through spoken and written words, symbols, and digital technology to share knowledge across generations.



2. The Three Vital Functions

Las tres funciones vitales que todo ser vivo realiza para mantenerse con vida



1. Nutrition

Función de Nutrición

Getting nutrients, water, and oxygen from our environment to obtain energy, grow, and repair our tissues.

BODY SYSTEMS INVOLVED:
Digestive • Respiratory • Circulatory • Excretory



2. Interaction

Función de Relación

Detecting changes and stimuli in the world around us (light, sound, danger) and responding quickly to survive and adapt.

BODY SYSTEMS INVOLVED:
Sense Organs • Nervous System • Locomotor (Bones & Muscles)



3. Reproduction

Función de Reproducción

The ability to produce offspring (children) that are similar to their parents, ensuring the continuation of the human species.

BODY SYSTEMS INVOLVED:
Male & Female Reproductive Systems • Hormones

3. The Cell: Building Block of Life

La célula: la unidad básica, estructural y viva más pequeña de todos los seres vivos

🔬 Definition of a Cell:

A cell is the **smallest living unit** capable of performing all three vital functions (nutrition, interaction, reproduction) on its own.

(La célula es la unidad con vida propia más pequeña que realiza nutrición, relación y reproducción).

🦠 Unicellular

Organismos Unicelulares

Made of just **one single cell**. They are microscopic.
Examples: Bacteria, yeasts, amoebas.

👤 Multicellular

Organismos Pluricelulares

Made of **millions or trillions of cells** working together in specialized teams.
Examples: Humans, trees, eagles, fish.

💡 HERO FACT!

¿SABÍAS QUE...?

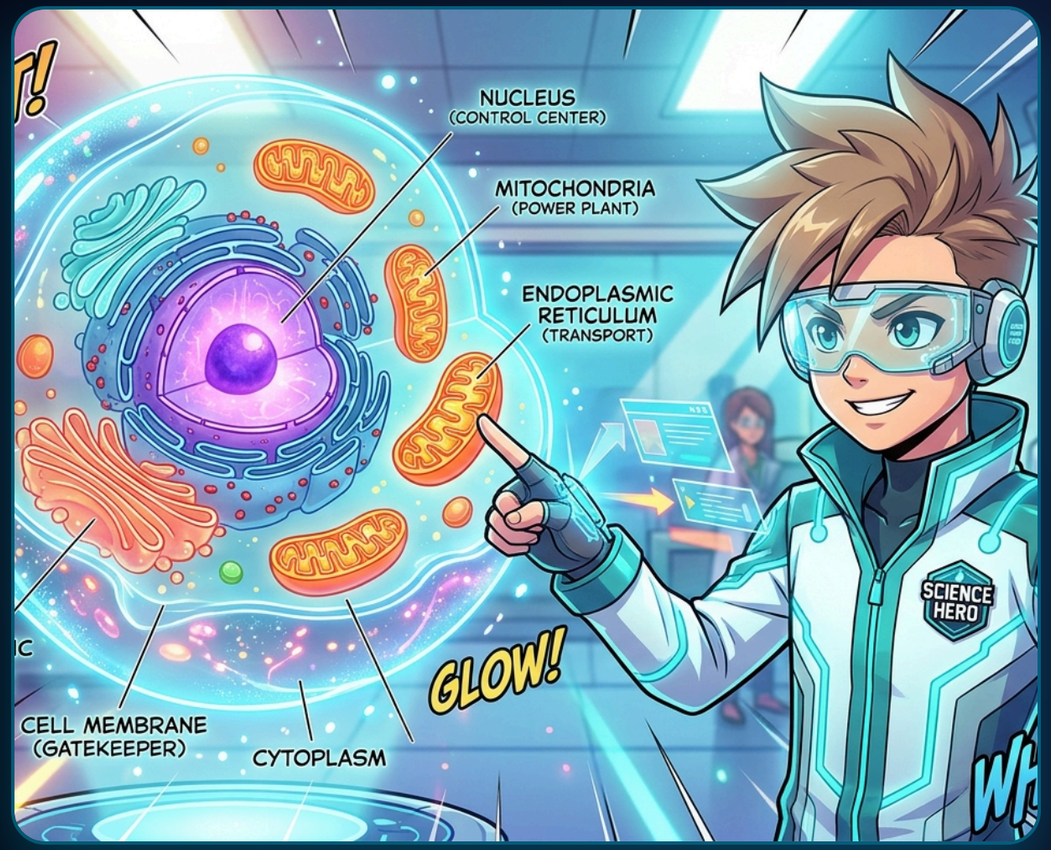
30+ Trillion

Your body contains more than **30,000,000,000,000 human cells**!

- **Red blood cells** carry oxygen like superhero courier ships.
- **Nerve cells (neurons)** send electric signals faster than lightning!

4. Inside the Animal Cell

Anatomía y orgánulos de la célula animal: los componentes esenciales que nos forman



1. **Cell Membrane** (Membrana Celular)
   The **protective outer boundary**. It regulates what enters (nutrients, water) and what leaves (waste products).

2. **Nucleus** (Núcleo)
   The **control centre** of the cell. It houses DNA (genetic instructions) and controls all cell actions and reproduction.

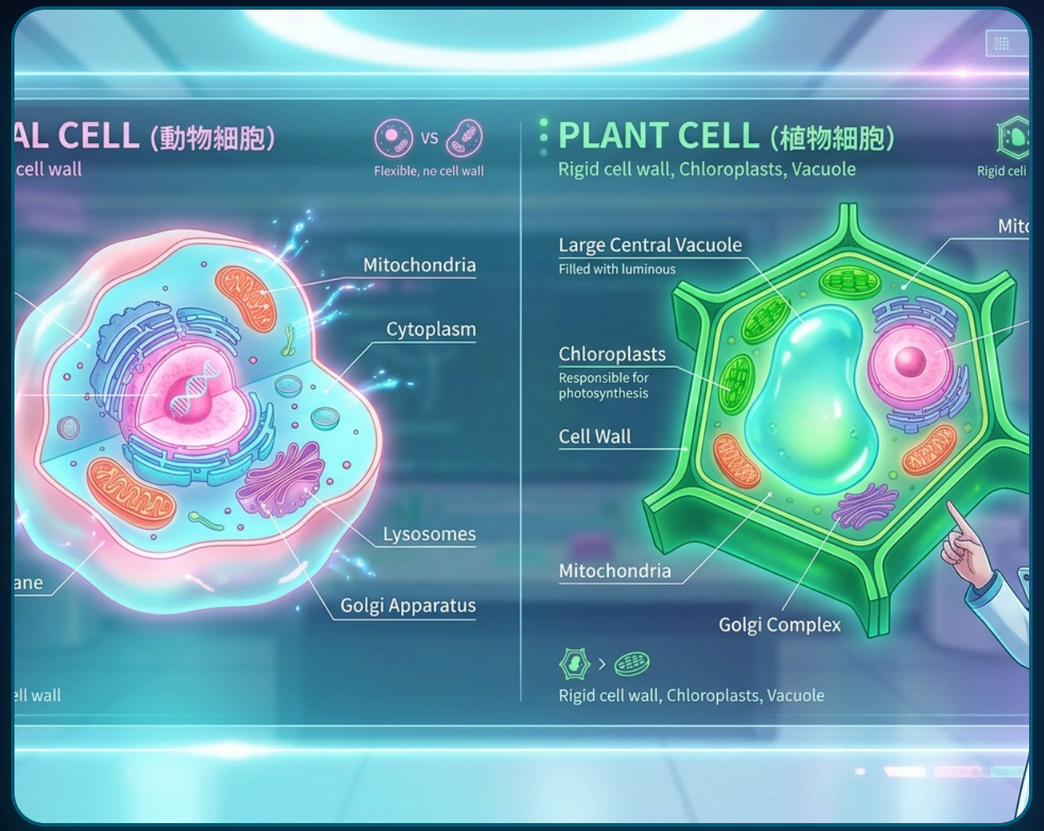
3. **Cytoplasm** (Citoplasma)
   A **clear, jelly-like liquid** that fills the cell. All organelles float in it and chemical reactions take place here.

4. **Mitochondria** (Mitocondrias)
   The **power plants** of the cell. Through cellular respiration, they turn glucose and oxygen into usable energy (ATP)!

5. **Vacuoles & Organelles** (Vacuolas y orgánulos)
   Small storage bubbles that hold water, nutrients, and waste materials until expelled.

5. Animal Cell vs. Plant Cell

Comparativa: ¿Qué tienen en común y qué orgánulos exclusivos poseen las plantas?



Animal Cell (Humans & Animals)

ROUND & FLEXIBLE

- Has **cell membrane, nucleus, cytoplasm, and mitochondria**.
- **No cell wall** (allowing animals to bend, flex, and move).
- Contains small, numerous vacuoles.

Plant Cell (Exclusive Features)

RIGID & GREEN

1. **Cell Wall (Pared Celular):** A tough outer layer made of cellulose that gives the plant rigid structural support.
2. **Chloroplasts (Cloroplastos):** Green solar powerhouses with chlorophyll that capture sunlight to perform photosynthesis.
3. **Large Central Vacuole (Gran Vacuola):** Stores huge amounts of water to keep the plant upright.

🔑 **Key Takeaway:** Human beings do NOT have chloroplasts or cell walls — that is why we cannot do photosynthesis and why our skin is flexible!

6. The 5 Levels of Organisation

De lo microscópico a lo macroscópico: cómo se ensambla el cuerpo humano paso a paso

1

Cell
CÉLULA

The basic living building unit.

Example:
Muscle cell
(Célula muscular)

2

Tissue
TEJIDO

A group of similar cells specialized in one task.

Example:
Muscle tissue
(Tejido muscular)

3

Organ
ÓRGANO

Different tissues working together for a purpose.

Example:
The Heart
(El corazón)

4

System
APARATO / SISTEMA

Multiple organs coordinated in harmony.

Example:
Circulatory system
(Aparato circulatorio)

5

Organism
ORGANISMO

The complete living human superhero!

Example:
Human being
(Ser humano)

7. Major Human Body Systems

Los principales aparatos y sistemas organizados según su función vital

🍎 Nutrition Systems

Sistemas para la Nutrición

- **Digestive System:** Breaks down food into nutrients.
- **Respiratory System:** Absorbs oxygen and expels $ ext{CO}_2$.
- **Circulatory System:** Blood transports nutrients & $ ext{O}_2$.
- **Excretory System:** Kidneys filter waste from blood.

⚡ Interaction Systems

Sistemas para la Relación

- **Sense Organs:** Eyes, ears, skin, tongue, nose collect input.
- **Nervous System:** Brain & nerves process data and give orders.
- **Locomotor System:** Skeleton (206 bones) and muscular system (over 600 muscles) produce motion.

🧬 Reproduction Systems

Aparatos de Reproducción

- **Male Reproductive:** Produces sperm in testicles.
- **Female Reproductive:** Produces ova in ovaries; carries baby in uterus.
- **Endocrine System:** Glands produce hormones that regulate growth and changes in puberty.

8. Stages of Human Life

Las etapas del desarrollo humano desde el nacimiento hasta la vejez



Infancy

0 - 3 AÑOS

Rapid growth, learning to walk, talk, and basic self-control. Complete dependence on parents.



Childhood

3 - 11 AÑOS (YOU!)

School years, active learning, motor skill refinement, making friends, baby teeth fall out!



Adolescence

11 - 18 AÑOS

Puberty, rapid growth spurts, physical changes, independence, identity formation.



Adulthood

18 - 65 AÑOS

Physical and emotional maturity, careers, building independent lives and starting families.



Old Age

65+ AÑOS

Senior years, retirement, sharing lifetime wisdom. Body requires extra care, exercise, and love.

9. Vocabulary Vault • Banco de Vocabulario

Términos esenciales en inglés con su traducción al castellano y definición clave

Cell
CÉLULA
The smallest living unit of all organisms.

Nucleus
NÚCLEO
Command centre containing DNA.

Cell Membrane
MEMBRANA CELULAR
Outer barrier controlling entry & exit.

Cytoplasm
CITOPLASMA
Gel-like fluid inside the cell.

Mitochondria
MITOCONDRIA
Organelles producing vital energy (ATP).

Tissue
TEJIDO
A group of similar, specialized cells.

Organ
ÓRGANO
Tissues working together (e.g. heart).

Organ System
APARATO / SISTEMA
Coordinated group of organs.

Organism
ORGANISMO
A complete, independent living being.

Opposable Thumbs
PULGARES OPONIBLES
Thumbs that can grip and manipulate tools.

Bipedalism
BIPEDESTACIÓN
Walking upright on two feet.

Vital Functions
FUNCIONES VITALES
Nutrition, Interaction & Reproduction.

10. Hero Science Mission Check!

Reto interactivo de aula para comprobar lo aprendido en la Unidad 1

❓ Mission 1

The Cell Battery

Which organelle acts as the power plant of human cells by producing energy from oxygen and nutrients?

ANSWER:
The Mitochondria! ⚡

❓ Mission 2

The 5 Steps Ladder

Put these levels in order from smallest to biggest:

Organ • Cell • Organism • Tissue • Organ System

ANSWER:
Cell → Tissue → Organ → System → Organism!

❓ Mission 3

Plant vs. Animal

Name two structures that plant cells have, but human cells do NOT have!

ANSWER:
Cell Wall & Chloroplasts! 🌿